DHARMSINH DESAI UNIVERSITY, NADIAD

FACULTY OF MANAGEMENT AND INFORMATION SCIENCE

B.C.A. SEMESTER VI

SUBJECT: (25BCA601)SYSTEM DEVELOPMENT PROJECT

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Project Definition: RestoFlow – A full-stack Restaurant Management Platform.

Company Timing: 11:00 AM to 6:00 PM

Week	Date	Day	Name of the Topic	Signature
Twelveth Week	16-02-2026	Monday	Feasibility study	
	17-02-2026	Tuesday	Test Cases created	
	18-01-2026	Wednesday	About the system(thesis)	
	19-01-2026	Thursday	Thesis creation	
	20-01-2026	Friday	Thesis creation	

External Guide Name: Somil Sir

Signature

Internal Guide Name: Prof. Rupal Patel

Signature

16-02-2025

1. Technical Feasibility

The Restaurant Management System has been successfully developed using a modern and scalable technology stack (ASP.NET Core, SQL Server, React, Stripe, SignalR).
All core modules — authentication, restaurant management, reservations, orders, payments, analytics, and real-time updates — are fully implemented and production-ready.
The system is secure, scalable, and optimized for performance with proper database design, cloud deployment readiness, and secure payment integration.
Status: Technically Sound and Successfully Implemented

2. Legal Feasibility

The system complies with data protection and payment security standards.

- Secure authentication and encrypted data handling implemented
- Stripe ensures PCI-DSS compliant payment processing
- Privacy policy, terms of service, and refund policies defined
- Designed to support regulatory and tax compliance requirements

Status: Legally Compliant and Safe for Deployment

3. Operational Feasibility

The system effectively supports real-world restaurant operations.

- Customers can order and reserve tables easily
- Restaurant admins can manage menus, orders, payments, and analytics
- System admins can manage users and restaurants centrally

The workflow aligns with standard restaurant processes, and minimal training is required for staff.
Status: Operationally Practical and User-Friendly

4. Schedule (Time) Feasibility

The project has been completed within the planned timeline.
All development phases — implementation, testing, optimization, and deployment preparation — were successfully executed.
The system is now ready for full-scale production use and future scalability.
Status: Completed Within Planned Timeframe

17-02-2026

1. Registration (Admin, RestaurantAdmin, Customer)

Sr No	Field	Test Data (input)	Expected Output	Actual Output	Result
1	Full Name*	Null	Full name is required	Username is Required	Failed to register
		1234	Enter alphabet only	Username is Required	Failes to register
		Himanshu Ladha	Register successful	Register successful	Register Success

2	Email*	Null	Email is required	Email is required	Failed to register
		*^@gmail.com	Invalid email	Invalid email	Failed to register
		himanshu@.com	Invalid email	Invalid email	Failed to register
		Himanshuladha8@gmail.com	Valid email	Valid email	Register Successful
3	Password*	Null	Password is required	Password is required	Failed to register
		him@1	Invalid Password	Password must be 6 char long	Failed to register
		him@123	Valid Password	Register successfully	Register success
4	Phone number*	Null	Number is required	Number is required	Failed to register
		123456789	Invalid Number	Number should be of 10 digit	Failed to register
		1234567890	Register Successfully	Register Successfully	Register Success
5	Role	Admin	Register Successfully	Register Successfully	Register Success
		Restaurant Admin	Register Successfully	Register Successfully	Register Success

Note : * field also used for Customer as well.

2. Login (Admin, RestaurantAdmin, Customer)

Sr No	Field	Test data	Expected Output	Actual Output	Result
1	Email	Null	Email is required	Email is required	Failed to Login
		*^@gmail.com	Invalid email	Invalid email	Failed to Login
		himanshu@.com	Invalid email	Invalid email	Failed to Login
		Himanshuladha876@gmail.com	Email doesn't match	Invalid email	Failed to Login
		himanshuladha8@gmail.com	Valid email	Valid email	Login Success
2	Password	Null	Password is required	Password is required	Failed to login
		him@1	Invalid Password	Password must be 6 char long	Failed to login
		him@12	Valid Password	Valid Password	Login Success
3	OTP	Null	OTP is required	OTP is required	Failed to login

		1234	Invalid OTP	Invalid OTP	Failed to login
		1222	Valid OTP	Valid OTP	Login Success

Note : Login field is same for Admin, RestaurantAdmin, Customer

3. Forgot Password

Sr no	Field	Test Data	Expected Output	Actual Output	Result
1	Email	Null	Email is required	Email is required	Failed to Login
		*^@gmail.com	Invalid email	Invalid email	Failed to Login
		himanshu@.com	Invalid email	Invalid email	Failed to Login
		Himanshuladha876@gmail.com	Email doesn't match	Invalid email	Failed to Login
		himanshuladha8@gmail.com	Valid email	Valid email	Login Success
2	New Password	Null	Password is required	Password is required	Failed to change
		him@1	Invalid Password	Password must be 6 char long	Failed to Change
		him@12	Valid Password	Valid Password	Enter the New password
3	Confirm New Password	Null	Password is required	Password is required	Failed to Change
		him@1	Invalid Password	Password should be same	Failed to Change
		him@12	Valid Password	Valid Password	Password changed successfully

18-02-2026

About the system

RestoFlow is a modern, QR code–based food ordering and restaurant management platform designed to simplify and enhance the dining experience. The system allows customers to scan a QR code placed on their table to access the digital menu of the respective restaurant. Through this interface, customers can conveniently browse menu items, place orders directly from their table, and complete secure online payments before the order is processed.

The platform supports real-time payment processing and enables customers to track the live status of their orders, ensuring transparency and improved customer satisfaction. By eliminating the need for physical menus and manual order-taking, RestoFlow enhances operational efficiency and reduces waiting time within the restaurant.

On the administrative side, the Restaurant Admin has full control over restaurant operations within the system. They can add, update, and delete food items, manage table information, and monitor customer orders efficiently. This ensures smooth day-to-day management of restaurant activities.

Additionally, the system includes a Main Admin module, which oversees the entire platform. The Main Admin can manage restaurants and customers, including adding, updating, and deleting restaurant records and food items. This centralized control ensures proper regulation, data management, and overall system administration.

Overall, RestoFlow integrates customer convenience with efficient restaurant management through a secure, real-time, and user-friendly digital platform.

The system leverage the uses of technologies like Asp.net 8 web core API (Backend) and React (Frontend) ensuing the smooth performance, scalability, and responsiveness across devices – Desktop, tablets and mobile phones. The integration of secure authentication and dynamic menus and payment makes this website a reliable and user friendly for all.

Objective:

The primary objective of this to make easy ordering from the restaurant with the help of mobile phones, offering easy features.

Types in this system:

1. Admin
2. Restaurant Admin
3. Customer/ User

19-02-2026

- Started creating the thesis on the basis of current data.

20-02-2026

- Continued the thesis creation.